

STRAMLIT FRONTEND:

How to Run the Penalty Kick Streamlit App

1. Manual Steps to Launch the App

1. **Open a terminal or command prompt.**
2. **Go to the folder where your app.py file is saved.**
 - For example:

text

`cd /path/to/your/project`

3. **Run the Streamlit app:**

`streamlit run app.py`

4. **A web browser window will open** and you will see the interactive penalty prediction interface.

2. How the Model Results Are Used

- **Biogeme results (parameters) are not loaded from a file at runtime.**
- Instead, all coefficients (from your previous naivemodel or Biogeme output file) have been manually copy-pasted into the Python function `get_sample_parameters()` inside `app.py`.
- When you use the app, these values are already set in the code and used for every prediction.

3. Data Files Are Not Needed

- **The app does NOT load or require any data file to run.**
 - No CSV, Excel, or other data files need to be present.
- All user inputs (like player foot, match situation, goalkeeper stats, etc.) are set via the sliders and selectboxes in the web interface.
- These inputs—alongside the hard-coded model coefficients—are used to calculate the prediction on-the-fly.

4. Summary

- **No dataset or model file needs to be loaded:** All needed numbers are inside the code.
- **You only need the app.py file and Streamlit installed** to run the app.
- **User chooses parameters on the web page** and gets instant model-based predictions.

VARIABLES IN SLIDE AND THEIR DESCRPTION:

Biogeme Parameter Prefix	App Variable Name	UI Widget Type	Description
b_foot_	foot	Selectbox (0/1)	Player's preferred shooting foot (Left=0, Right=1)
b_moveGK_	moveGK	Selectbox (0/1)	Whether goalkeeper moves early before shot (No=0, Yes=1)
b_IS_	IngSo	Selectbox (0/1)	Is penalty part of a shootout? (No=0, Yes=1)
b_comL_	comLea	Selectbox (0/1)	Competition level (Non-League=0, League=1)
b_HGK_	HeiGK	Selectbox (1-16)	Goalkeeper height coded (1:<183 cm, 16:>196 cm, with increments as per thesis coding)
b_dec_	Dec	Selectbox (0/1)	Decisive penalty? (No=0, Yes=1)
b_lp_	lpbg	Selectbox (0/1/2)	Last penalty result (0=N/A, 1=Failed, 2=Scored)
b_latr_	la3	Selectbox (0/1/2)	Last shot to alternative 3 (Top Right) (0=N/A,1=No,2=Yes)
b_ladr_	la6	Selectbox (0/1/2)	Last shot to alternative 6 (Down Right) (0=N/A,1=No,2=Yes)
b_perc	perc	Slider (0.0-1.0)	Overall historical success percentage (percentage of previous penalties scored)
b_rnog_	rnoGroup	Selectbox (0/1)	Is match in a group stage? (No=0, Yes=1)
b_solsbg_	Solsbg	Selectbox (0/1)	Opponent's last shootout penalty success (Failed=0, Scored=1)
b_def_	def_var	Selectbox (0/1)	Player position: Defender? (No=0, Yes=1)
b_nmovGK_	nmovGK	Selectbox (0/1)	Goalkeeper movement status during run-up (Moving=0, Not moving=1)